Metric Traveling Salesperson Problem: Exhaustive Search and Greedy Heuristics

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GitHub Repository: `https://github.com/AbdullahKhan5880/Metric-TSP-Project`

***Abstract* – The Traveling Salesperson Problem (TSP) is a classical combinatorial optimization problem in which the objective is to find a minimum-cost Hamiltonian cycle in a weighted graph. This project investigates two approaches: exhaustive search and a deterministic multi-start nearest-neighbour greedy heuristic. Both algorithms were implemented in Python and evaluated on handmade and synthetic undirected graphs. The experimental study considered four edge-density levels (12.5%, 25%, 50%, and 75%) and graph sizes ranging from 4 to 16 vertices. A total of 28 synthetic instances and 56 solver executions were recorded. Under a 90-second time limit, the exhaustive algorithm completed all density experiments up to 12 vertices, while all experiments at 14 and 16 vertices reached the timeout. The results illustrate the rapid growth of exhaustive search and the substantially lower execution time of the greedy method. The limitations of applying metric-TSP terminology to sparse Euclidean subgraphs are also discussed.**

***Keywords* – Traveling Salesperson Problem, Exhaustive Search, Greedy Heuristics, Computational Complexity, Graph Algorithms**

I. Introduction

The Traveling Salesperson Problem (TSP) is a well-known combinatorial optimization problem. Given a set of vertices and weighted edges, the goal is to find a minimum-cost tour that visits every vertex exactly once and returns to its starting point [1].

The Metric TSP is a special form in which distances satisfy non-negativity, symmetry, and the triangle inequality. It has applications in transportation, manufacturing, logistics, and engineering design.

The objective of this project is to implement and compare two algorithmic approaches: exhaustive search and a deterministic greedy heuristic. The comparison focuses on execution time, operation counts, solution feasibility, and solution quality.

II. Problem Formulation and Applications

A. Mathematical Formulation

Let an undirected weighted graph be represented by

$$G = (V, E, w),$$

where V is the set of vertices, E is the set of edges, and $w : E \to \mathbb{R}^+$ assigns a positive weight to each edge.

For a complete metric graph, the distance function also satisfies the triangle inequality:

$$w(u, v) \leq w(u, x) + w(x, v).$$

A Hamiltonian cycle is a sequence

$$(v_0, v_1, \ldots, v_{n-1}, v_0)$$

that visits every vertex exactly once before returning to its starting vertex. Every consecutive pair must be connected by an edge.

The objective is to minimize the total tour cost:

$$\min \sum_{i=0}^{n-1} w(v_i, v_{(i+1) \bmod n}).$$

If no Hamiltonian cycle exists, the algorithm reports that no tour exists when this has been established by exhaustive search. A heuristic failure, however, does not prove infeasibility.

B. Real-World Applications

One relevant application is robotic path planning in computational engineering. A robot may need to visit a collection of inspection locations in a manufacturing facility while minimizing total travel distance. A tour can represent a candidate inspection route.

A second application is the scheduling of measurement or inspection operations in engineering laboratories. Equipment may need to visit multiple measurement positions, and minimizing travel distance can reduce operational time.

These applications motivate the study of efficient algorithms for route optimization.

III. Algorithm Design

A. Exhaustive Search

The exhaustive search algorithm evaluates candidate vertex permutations to identify a minimum-cost Hamiltonian cycle.

To avoid equivalent representations of an undirected cycle, the algorithm fixes the first vertex and eliminates reverse-direction duplicates. For each remaining candidate, it checks whether all required edges exist and calculates the total cost when a valid cycle is found.

The algorithm returns an optimal tour when the search completes. It can also report that no tour exists after examining all candidates. A time limit is used to prevent excessively long executions.

The number of candidate permutations grows factorially. Its asymptotic complexity is approximately

$$O((n-1)!),$$

with additional work required to verify the edges of each candidate. The TSP is a classical computationally difficult optimization problem [2].

B. Greedy Nearest-Neighbour Heuristic

The greedy algorithm constructs a tour by repeatedly selecting the nearest unvisited neighbour of the current vertex.

The implementation uses a multi-start strategy: it attempts construction from each vertex and retains the lowest-cost feasible tour found. Ties are resolved deterministically according to vertex insertion order.

Nearest-neighbour methods are established heuristic approaches for the TSP, although they do not generally guarantee an optimal tour [3].

Furthermore, a greedy construction may fail to find a Hamiltonian cycle even when one exists.

For the implemented multi-start version, the worst-case time complexity is

$$O(n^3).$$

IV. Experimental Methodology

A. Implementation and Test Instances

The algorithms were implemented in Python using NetworkX for graph representation and supporting graph operations.

Two handmade graphs, with 5 and 6 vertices, were used to validate the correct behaviour of both algorithms. These instances were considered separately and were not included in the performance evaluation. The benchmark dataset consisted of 28 synthetic graphs, generated using different numbers of vertices and edge densities. Each graph was evaluated using both algorithms, resulting in 56 solver executions in total.

Synthetic graphs were generated using distinct two-dimensional integer coordinates in the interval $[1, 500]$. Edge weights were calculated using Euclidean distance. The student identifier, 138367, was used as the random seed.

The experiments considered:

- Vertex counts: 4, 6, 8, 10, 12, 14, and 16.
- Edge densities: 12.5%, 25%, 50%, and 75%.
- Number of synthetic instances: 28.
- Maximum execution time per exact run: 90 seconds.

Both algorithms were executed on the same generated instances. Execution time was measured using Python's `time.perf_counter()`.

B. Evaluation Metrics

The evaluation considered execution time, basic operations, tested configurations, solution quality, and algorithm status.

When both algorithms produced comparable feasible solutions and the exact optimum was known, the optimality gap was calculated as

$$\text{Gap}(\%) = \frac{|S_g - S^*|}{S^*} \times 100,$$

where S_g is the greedy solution cost and S^* is the optimal cost.

The operation counters are implementation-specific: exhaustive search counts edge-existence checks, whereas greedy search counts candidate inspections and return-edge checks. Consequently, these counts are useful for studying each implementation but are not strictly equivalent measures of computational work.

V. Results and Discussion

A. Benchmark Overview

The benchmark sweep produced 56 execution records from 28 graph instances. The exhaustive solver reported 7 optimal solutions, 13 cases with no Hamiltonian cycle, and 8 timeouts. The greedy solver reported 6 feasible tours and 22 unsuccessful constructions.

These statuses require careful interpretation. An exhaustive `no_tour` result proves that no Hamiltonian cycle exists for that instance, because all candidates were examined. A greedy `no_tour` result only means that the heuristic did not construct a tour; it does not prove infeasibility. A timeout means that the exact optimum remains unknown.

B. Execution Time

Table I presents mean execution times across the four density levels for each vertex count.

TABLE I
Mean execution time in seconds

Vertices	Exhaustive	Greedy
4	0.000064	0.000061
6	0.000356	0.000140
8	0.007730	0.000149
10	0.395589	0.000291
12	42.484927	0.000402
14	`timeout (90 s)`	0.000728
16	`timeout (90 s)`	0.000936

Exhaustive-search runs that reached the 90-second time limit are reported as timeouts rather than completed execution times.

Figure 1 presents the execution-time comparison using a linear scale.

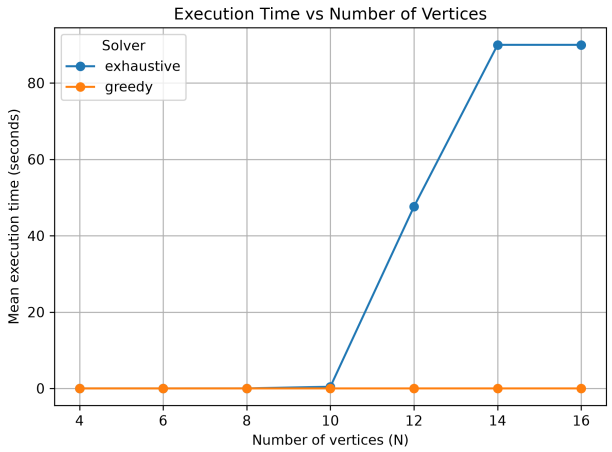


Fig. 1 - Execution time versus number of vertices (linear scale).

Figure 2 presents the same comparison using a logarithmic scale, making the differences at smaller execution times easier to observe.

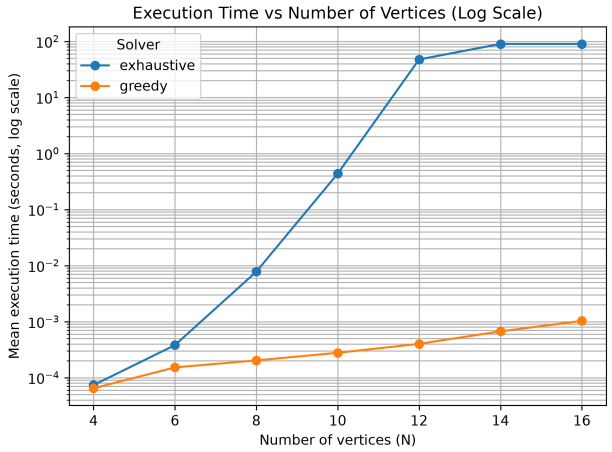


Fig. 2 - Execution time versus number of vertices (logarithmic scale).

The exhaustive algorithm exhibits a rapid increase in execution time as the number of vertices grows. At 12 vertices, its mean execution time was approximately 42.48 seconds. At 14 and 16 vertices, all four density experiments reached the 90-second timeout.

In contrast, the greedy algorithm remained below one millisecond on average for all tested sizes. This demonstrates its practical speed advantage, although execution speed alone does not establish solution quality.

C. Basic Operations

Table II presents the mean operation counts for both algorithms.

The operation counters are implementation-specific and should not be interpreted as directly comparable measures of work. For exhaustive search, the counter records edge-existence checks, whereas for the greedy algorithm it records candidate inspections and return-edge checks. Therefore, the values are mainly useful for observing the growth of operations within each algorithm.

TABLE II
MEAN BASIC OPERATIONS

Vertices	Exhaustive	Greedy
4	7	20.5
6	109	72.5
8	3,759	194
10	290,380	404
12	33,380,458	705.5
14	87,970,806	1,199.5
16	86,642,205	1,784.5

Figure 3 illustrates the operation-count comparison using a logarithmic scale.

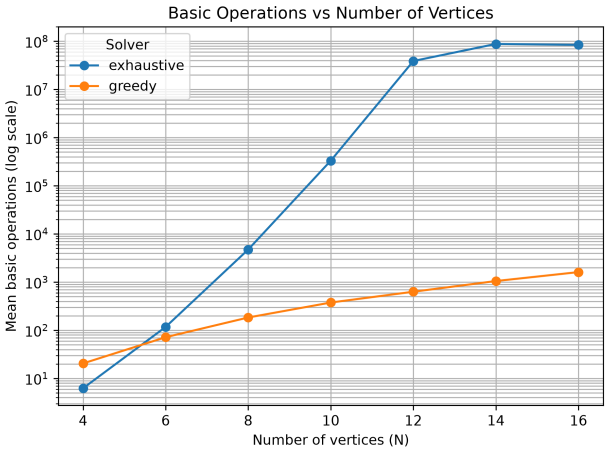


Fig. 3 - Mean basic operations versus number of vertices.

The exhaustive operation count increases substantially with graph size. Counts at 14 and 16 vertices represent partial work before timeout, rather than completed searches. The greedy operation count grows much more gradually.

D. Optimality Gap

Only two benchmark cases provided a valid comparison between a completed exact solution and a feasible greedy solution.

For six vertices at 75% density, the greedy solution matched the optimum, producing a gap of 0%. For ten vertices at 75% density, the gap was approximately 5.18%.

Figure 4 shows the available greedy optimality-gap measurements.

These two observations are insufficient to establish a general quality guarantee. More instances with known optimal solutions would be needed for a statistically meaningful comparison.

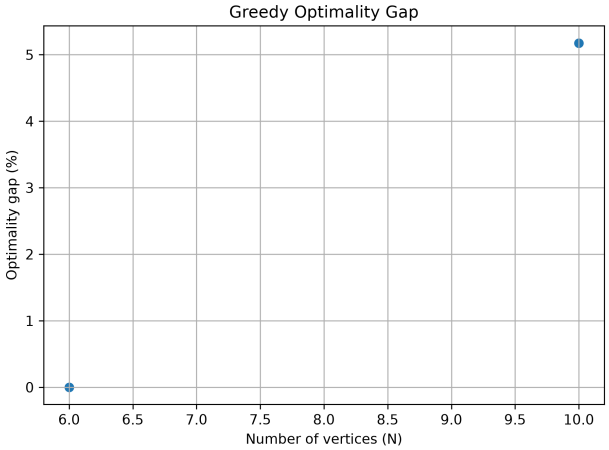


Fig. 4 - Optimality gap of feasible greedy solutions.

E. Maximum Solvable Instance Size

Under the 90-second limit, the largest tested vertex count for which all four density experiments completed using exhaustive search was

$$N_{\max} = 12.$$

All exact runs at 14 and 16 vertices reached the timeout. This empirical limit is consistent with the factorial growth of exhaustive search. Larger instances would require more efficient exact methods or approximation techniques.

VI. Limitations

An important modelling limitation concerns graph density. The synthetic instances were generated as sparse subgraphs of Euclidean point sets. Although the edge weights are Euclidean distances, the underlying Euclidean distance function satisfies the triangle inequality for all pairs of points. However, the generated graphs contain only a subset of the possible edges. Therefore, they are not complete metric-TSP instances under the standard formulation. The experiments should be interpreted as Hamiltonian-cycle optimization on sparse graphs with Euclidean edge weights.

The limited number of instances with both known exact and greedy solutions also restricts conclusions about heuristic solution quality. In addition, the greedy method's failure to construct a tour should not be interpreted as proof that no tour exists.

VII. Conclusion

This project implemented and evaluated exhaustive search and a deterministic multi-start nearest-neighbour heuristic for the Traveling Salesperson Problem.

The experiments demonstrate the factorial growth of exhaustive search and the substantially lower execution time of the greedy approach. Under a 90-second limit, the largest tested size for which all four density experiments completed exactly was 12 vertices.

Although the greedy method was computationally efficient, it did not always construct a feasible tour and provided no optimality guarantee. The small number of comparable solutions also limits conclusions about its approximation quality.

Future work in Project 2 can investigate refined heuristics, local search, and randomized approximation strategies to improve solution feasibility and quality on larger instances.

Generative AI Declaration

I used AI as a supporting tool throughout this project to clarify algorithmic concepts, understand theoretical aspects, troubleshoot programming issues, and improve the organization and presentation of the report. I independently implemented the algorithms, executed the experiments, collected and verified the results, and analysed the findings.

References

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[3] Daniel J. Rosenkrantz, Richard E. Stearns, and Philip M. Lewis II, "An analysis of several heuristics for the traveling salesman problem", *SIAM Journal on Computing*, vol. 6, no. 3, pp. 563–581, 1977.